

Optimizing Transfer Value in Humanitarian Food Assistance: Welfare and Community Impacts from the DRC

Jonathan Garcia* Aysha Twose†

June 4, 2026

Abstract

This paper studies how the design of transfer values in humanitarian food assistance affects household welfare, intra-household dynamics, and community outcomes. We exploit a pilot program with three sources of variation across two modalities: vulnerability-differentiated food rations (2,100 vs. 1,260 kcal per capita per day), a 33% increase in cash transfers, and a shift from per capita to economies-of-scale-adjusted allocations. Using experimental, quasi-experimental, and qualitative data, we find that a 40% reduction in food rations generates disproportionately large welfare losses—including a 44% decline in dietary diversity—consistent with households operating near consumption thresholds. These costs are borne primarily by women, while children’s nutritional outcomes remain unchanged. Cash transfers outperform in-kind assistance across food security and expenditure outcomes, and non-uniform in-kind transfers raise perceived tensions, while the same variation in cash has no effect. Adjusting transfer values for economies of scale yields comparable welfare at lower cost, enabling a 2.2% expansion in coverage.

Keywords: humanitarian food assistance, transfer design, social cohesion

JEL Codes: O12, I38, D12, D13, Q18

*Zero Hunger Lab, Humboldt University of Berlin; Leibniz Institute of Vegetable and Ornamental Crops (IGZ). Corresponding author: jonathan.garcia@hu-berlin.de.

†World Food Programme.

1 Introduction

The number of people experiencing hunger and in need of humanitarian food assistance (HFA) has consistently exceeded available resources, but this imbalance has recently intensified. In 2024, an estimated 281.6 million people faced high levels of acute food insecurity (GRFC, 2024), while the global humanitarian response plan faced a funding gap of 75%, equivalent to USD 37 billion (OCHA, 2024). This structural shortfall has been further exacerbated by recent contractions in aid budgets, including the dismantling of the United States Agency for International Development (USAID) and reductions by other major donors. As a result, policymakers are increasingly required to optimize the allocation of limited resources. To date, most of these adjustments have occurred along the extensive margin, primarily through reductions in the number of programs or beneficiaries served. Yet, little attention has been devoted to optimization along the intensive margin, namely how much assistance to provide to each recipient, how transfer values are determined, and the implications of these choices, even though transfer size represents one of the largest cost components of these programs.

Most of the existing evidence on optimizing the design of assistance programs focuses on alternatives in targeting, modality (see, for instance, Gentilini et al., 2017; Bastagli et al., 2016), frequency (see Haushofer and Shapiro, 2016), and, more recently, the timing of assistance (see Brune et al., 2016; Fink, Jack, and Masiye, 2020). Evidence on how outcomes respond to variation in transfer size remains limited and mixed, suggesting that the shape of returns to transfer size appears to depend on context, population, and outcome of interest, and remains insufficiently understood, particularly in humanitarian settings, including how households and communities respond to reductions in transfer values or to variation in transfer size across households with similar characteristics.

To contribute evidence on these gaps, we exploit three variations in the determination of transfer values introduced by the World Food Programme (WFP) in the Kasai region of the Democratic Republic of the Congo (DRC), where transfer values are traditionally determined on a per capita basis, prorated by household size, and identical across households of the same size deemed food insecure. While operationally simple, this allocation

rule may generate inefficiencies in the presence of economies of scale within households or when needs and vulnerabilities vary across households (ref), potentially leaving resources that could be reallocated to expand horizontal coverage without compromising welfare outcomes.

The first variation involved a shift from uniform in-kind food assistance to transfers differentiated according to beneficiaries' levels of need. Specifically, WFP randomly assigned communities to receive either a uniform "full food basket," equivalent to 2,100 kcal per capita per day, or a differentiated transfer based on a proxy means test that classified households as either extremely food insecure or food insecure. Households classified as food insecure received an in-kind ration 40% lower than that of extremely food insecure households, hereafter referred to as the "half ration," reflecting the estimated average food gap required for these households to reach the minimum acceptable consumption threshold. The second and third variations were implemented for cash-based modalities. The second variation increased the transfer value to include a non-food component, calculated on a per capita basis and prorated by household size. This design compares households receiving a transfer equivalent in value to the in-kind full food basket with those receiving an augmented transfer that includes both food and non-food components. The third variation further modified the cash transfer design by incorporating the same non-food component while accounting for potential economies of scale within households, so that transfer values were no longer strictly proportional to household size.

We leverage the variation in transfer size using both experimental and quasi-experimental methods. We exploit the random assignment at the food distribution point level to examine how uniform versus non-uniform transfer sizes affect community dynamics across modalities. We find that distributing non-uniform rations within communities increases perceived conflict by 14 percentage points in in-kind villages, while the same variation delivered in cash has no effect on tensions or social cohesion, consistent with the visibility of in-kind distributions driving social frictions.

We then examine the impact of variation in in-kind rations and cash transfer values using a difference-in-differences approach. We find that a 40% reduction in food rations

generates disproportionately large welfare losses, with dietary diversity declining by 44%, food expenditure by 31%, and education spending by 67%, alongside sharp reductions in the consumption of nutrient-rich food groups. These costs are primarily absorbed by women, whose dietary diversity declines significantly in both modalities, while children’s nutritional outcomes remain unchanged, consistent with intra-household allocation patterns that protect children at the expense of women. Cash transfers substantially outperform in-kind assistance across all food security and expenditure measures (+0.45 SD in FCS and +0.96 SD in dietary diversity). Finally, adjusting transfer values to account for household economies of scale delivers comparable welfare outcomes at 2.2% lower cost, suggesting scope to expand coverage without compromising effectiveness.

These findings contribute to four strands of literature. First, we add to the limited evidence on the welfare effects of transfer value reductions in humanitarian settings. Much of the existing literature on transfer size focuses on large cash transfers in development settings, examining poverty traps and the marginal returns to increases in support (Balboni et al., 2022; Banerjee et al., 2019; Haushofer and Shapiro, 2016). Within this literature, evidence remains mixed, with some studies documenting diminishing marginal returns (Filmer and Schady, 2014), others highlighting threshold effects driven by lumpy investments or behavioral responses (McIntosh and Zeitlin, 2021; Kondylis and Loeser, 2021), and recent meta-analyses suggesting that impacts may, on average, scale approximately proportionally with transfer size (Crosta et al., 2023, 2024). By contrast, this paper contributes evidence from the opposite margin—transfer *cuts*—in a context where beneficiaries are closer to subsistence and coping margins are thinner. The results show welfare losses that exceed proportional benchmarks, consistent with threshold effects near subsistence, a mechanism largely absent from the existing evidence base.

Second, we contribute evidence on aid discontinuation, which comes primarily from simulation studies in humanitarian contexts (WFP, 2023a, 2023b, 2023c) or from high-income countries, where terminating cash transfers has been linked to adverse effects on children’s human capital and increased criminal activity (Dustmann, Landersø, and Andersen, 2024). Instead, by studying calibrated reductions rather than outright dis-

continuation, we provide evidence on an intermediate policy option—one that preserves coverage while adjusting transfer intensity—whose welfare costs and social implications had not been previously quantified.

Third, we contribute to the literature on the social effects of non-uniform assistance within communities. The introduction and selective inclusion into programs have been linked to heightened tensions, perceptions of unfairness, and jealousy (Della Guardia et al., 2022; Paler et al., 2019; Premand and Schnitzer, 2021), and the removal of beneficiaries has been shown to damage social cohesion (Olivier et al., 2025). We extend this evidence by showing that the *modality* of delivery mediates the social cost: non-uniform in-kind transfers generate tensions, while non-uniform cash transfers do not. This finding has direct implications for how agencies implement budget-driven adjustments to transfer values, and suggests that the shift to cash-based programming may yield social benefits beyond the well-documented gains in cost-efficiency and household welfare.

Lastly, we advance the literature on the economic efficiency of aid allocation, which has primarily focused on targeting methods and their ability to minimize inclusion and exclusion errors while holding budgets constant (Verme and Gigliarano, 2019). We evaluate three design margins jointly—needs-based calibration, modality choice, and economies of scale—assessing each in terms of both welfare effects and cost implications. The results reveal a stark asymmetry: the modality switch and equivalence-scale adjustments offer substantial efficiency gains without welfare losses, while needs-based ration calibration imposes welfare costs that exceed the fiscal value of the resources freed. This suggests that not all optimization margins are created equal, and that the order in which agencies pursue them matters: gains from switching modality and adjusting for household size should be exhausted before reductions in transfer value are considered. The paper closes with program design simulations that quantify the coverage expansion achievable under each margin, providing agencies with actionable estimates for resource allocation decisions.

The rest of the paper is organized as follows. Section 2 describes the DRC context and WFP’s humanitarian assistance program, including the targeting and transfer

value determination process. Section 3 presents the research design, covering the transfer interventions, study design, and randomization. Section 4 develops the conceptual framework linking each source of variation to a testable prediction. Section 5 describes the data and empirical strategy. Section 6 presents the results, organized by research question: community tensions, household-level welfare effects, intra-household impacts, heterogeneities, economies of scale, modality comparison, program design simulations, and qualitative findings. Section 7 concludes.

2 Context and Program

2.1 Context

The Democratic Republic of the Congo (DRC) is the largest country in sub-Saharan Africa and is endowed with vast natural resources. Yet despite this potential, it ranked 180th out of 193 countries on the 2023 Human Development Index (UNDP, 2024), and consistently performs among the lowest across a range of development indicators such as life expectancy (60 years), education (e.g., only 25% of children complete primary school), and health outcomes (e.g., maternal mortality remains above 500 deaths per 100,000 live births). The country is facing a deepening humanitarian emergency, with an estimated 27.7 million people experiencing high levels of acute food insecurity (IPC, 2025). This includes nearly 3.9 million people in emergency situations and in need of humanitarian food assistance.

The situation of internally displaced persons (IDPs) in the DRC further illustrates the severity of the crisis. As of early 2025, more than 7.8 million people were displaced, most of them residing with host families in the conflict-affected eastern provinces. Among the 3.7 million IDPs analyzed in the most recent IPC assessment, over 2.2 million were facing high levels of acute food insecurity, including more than 738,000 in Emergency conditions. Large displacement in the country due to conflict and climate shocks has placed significant pressure on resident households, many of which were already facing severe economic hardship and food insecurity themselves. As a result, both displaced

and host communities are now contending with constrained access to food, livelihoods, and basic services, underscoring the urgent need for targeted humanitarian assistance and improved protection mechanisms.

The central provinces of Kasai, Kasai-Central, and Kasai-Oriental are among the areas most affected by the issues described above, which continue to generate significant humanitarian challenges. Armed conflict remains a primary driver of food insecurity, directly displacing millions, disrupting access to food and livelihoods, and constraining both humanitarian access and return movements that further strain host communities. In response, organizations such as WFP are increasingly integrating conflict-sensitive approaches into the design and delivery of their programs. This study focuses on four sites within the Kasai region: Kalemie and Kitangwa, where WFP delivers in-kind food assistance, and Mutema and Tshilenge, where assistance is provided through cash-based transfers. These sites were selected because they span both transfer modalities and cover areas with high levels of vulnerability, displacement, and food insecurity.

Country-specific evidence from the DRC on humanitarian assistance and cash transfer programs remains limited. A few studies have assessed the introduction of cash or value voucher interventions (e.g., Aker, 2017; Bonilla et al., 2017; Quattrochi et al., 2020), but there is little insight into how to prioritize beneficiaries or evaluate the effects of such prioritization at the community and household levels. In the context of a persistent hunger emergency, tensions, and aid cuts, evidence on how to implement aid cuts in the most conflict-sensitive manner is of high importance.

2.2 WFP’s Humanitarian Assistance

In 2021, the WFP in DRC provided humanitarian food assistance to nearly six million people in response to widespread food insecurity driven by conflict, displacement, and economic hardship. Of these beneficiaries, approximately 4.5 million received in-kind food assistance, while an additional 1.4 million were supported through cash transfers. By 2023, the number of beneficiaries declined to 5.3 million, reflecting both funding constraints and evolving needs across regions.

This study focuses on the Kasai Region, which includes the provinces of Kasai, Kasai Central, and Kasai Oriental, and pilots adjustments to the transfer value within WFP’s unconditional cash transfer program. The intervention targets highly vulnerable households, including internally displaced persons and host communities experiencing severe food insecurity. The objective is to examine how different transfer values influence key outcomes related to food security and household welfare in a setting marked by persistent vulnerability and limited resources.

Households participating in the program typically receive assistance once per year, with support provided over a six-month period through monthly installments. Depending on operational feasibility and local market conditions, the assistance is delivered either in cash or as in-kind food. This structured and time-limited modality is intended to stabilize consumption during periods of heightened need and to reduce reliance on harmful coping strategies, while also generating evidence on the effectiveness of varying cash transfer values.

2.3 Targeting and Transfer Value

WFP’s country office conducts regular targeting exercises to identify and support the most vulnerable households. The process begins by identifying geographic areas with high levels of food insecurity using data from the Emergency Food Security Assessment (EFSA), a standardized survey that collects household-level data on food consumption, coping strategies, livelihoods, and expenditure patterns. Within these areas, WFP applies a proxy means test (PMT) model, calibrated on EFSA data, to classify households by vulnerability. The PMT predicts the Food Consumption Score (FCS) for each household based on observable characteristics and assigns a binary classification: food insecure (eligible for assistance) or food secure (not eligible). This classification determines which households are included in the program. Complementary insights from focus group discussions conducted in target communities inform the operational design.

Food rations for eligible households are estimated using nutritional benchmarks designed to meet a daily intake of approximately 2,100 kilocalories per person. For this

evaluation, a standard full ration includes 400 grams of cereals, 120 grams of beans or split peas, 30 grams of oil, and 5 grams of salt, totaling 555 grams per person per day. This composition reflects guidance under the Sphere Standards, which outline minimum requirements for both macro and micronutrients in emergency settings, and aligns with donor standards such as those of USAID.

In the case of cash transfers, the transfer value is estimated using the Minimum Expenditure Basket (MEB), which represents the cost of a set of essential goods and services required to meet basic needs over a given period. The MEB typically includes food items, as well as essential non-food components such as hygiene products, cooking fuel, and basic health and education expenses, depending on the context. It is calculated based on regular market price monitoring and adjusted periodically to reflect inflation, seasonal changes, and evolving household needs. By anchoring the cash transfer value to the MEB, WFP aims to ensure that beneficiaries receive sufficient support to access the minimum resources needed to sustain themselves and their families, while preserving dignity and promoting local market engagement.

3 Research Design

3.1 Transfer Intervention

Food rations calibration. We leverage WFP’s standard targeting approach that uses a PMT model to classify households as food insecure or food secure based on a predicted FCS score. We move beyond the binary classification to distinguish two levels of vulnerability among beneficiaries: *extremely vulnerable* households, which face the most severe food insecurity and continue to receive the full food ration, and *vulnerable* households, which are food insecure but retain some capacity to meet a portion of their caloric needs. For both groups, the transfer value was determined by estimating the average gap between what households can afford and the cost of a minimum essential food basket. The minimum essential food basket was estimated through a four-step process:

- i)* **Estimation of household capacity.** Households engaging in negative coping

strategies that artificially increase income—such as distress asset sales or child labor—were excluded from the analysis. Among the remaining households, the average per capita expenditure was calculated for the second and third quintiles (20th–39th percentile), who were considered moderately vulnerable but not among the most food-insecure. This average was estimated at 13,921 CDF per person.

ii) Food share of total expenditure. Using household expenditure data, it was estimated that 70% of total spending was allocated to food. This proportion is consistent with the consumption patterns of poor households in contexts of high food insecurity.

iii) Estimation of food expenditure capacity. Applying the food expenditure share to the average per capita expenditure yielded an estimated capacity of 9,744 CDF per person for food-related purchases.

iv) Comparison to cost of a full food basket. The average cost of a full food basket sufficient to meet caloric requirements was estimated at 25,768 CDF. Dividing households' estimated food expenditure capacity by the cost of the basket indicated that poor households could afford approximately 38% of their caloric needs. Based on this analysis, the transfer value for vulnerable households was calibrated to cover the remaining 60% of caloric needs, resulting in a ration approximately 40% lower than the full ration provided to extremely vulnerable households.

Cash transfer expansion. The standard cash transfer (USD 14 per person per month) was designed to cover food needs only. To test whether covering non-food expenditures improves welfare outcomes, WFP introduced an expanded transfer of USD 18 per person per month—a 33% increase. The additional USD 4 per person was estimated based on the updated Minimum Expenditure Basket (MEB) from the DRC Cash Working Group,¹ which quantifies the cost of essential non-food items required to maintain a minimum

¹The Cash Working Group is a coordination body comprising WFP, UNHCR, and other humanitarian actors that maintains the MEB—the estimated cost of essential goods and services (food, hygiene, health, education, cooking fuel) required to meet basic needs. The MEB is updated periodically based on market price monitoring and reflects local cost-of-living conditions.

standard of living.

Transfer value adjusted for economies of scale. Under the standard approach, cash transfer values are *prorated*: the per capita amount is multiplied by household size ($T = c \times n$), so that a household of six receives exactly six times the transfer of a single-person household. This linear scaling does not account for the fact that larger households can share certain costs—shelter, cooking fuel, and some food preparation—that do not rise proportionally with household size. The economies-of-scale (EE) variation adjusts the transfer amount using an equivalence-scale formula ($T = c \times n^\beta$, with $\beta = 0.7$), aiming to improve adequacy across households with varying compositions while potentially freeing resources to expand program coverage.

3.2 Study Design

The study employs a mixed-methods design that combines experimental and quasi-experimental methods with qualitative data from focus group discussions, community consultations, and feedback mechanisms. The quantitative component exploits a randomized design implemented at the food distribution point (FDP) level within two modalities: in-kind food assistance and cash-based transfers. FDPs serve as the primary administrative unit for WFP’s program delivery, typically covering one or more villages. Randomization was stratified by modality and site, ensuring balance in baseline characteristics across treatment arms within each program area.

Unit of randomization and assignment. Within the in-kind modality (Kalemie and Kitangwa sites), FDPs were randomly assigned to either a *uniform* arm, where all households received the full food ration (2,100 kcal/person/day), or a *differentiated* arm, where households classified as extremely vulnerable received the full ration while those classified as vulnerable received a calibrated ration approximately 40% lower (1,260 kcal/person/day). Vulnerability classification was determined by a proxy means test (PMT) administered prior to randomization.

Within the cash modality (Mutena and Tshilenge sites), FDPs were randomly assigned

to one of three arms: (i) a standard transfer of USD 14 per person per month, covering food needs only; (ii) an expanded transfer of USD 18 per person per month, which added USD 4 for non-food expenditures; or (iii) an economies-of-scale (EE) transfer, where the transfer value was adjusted by household size using an equivalence-scale formula ($T = c \times n^\beta$, with $\beta = 0.7$), designed to account for the declining marginal cost of consumption in larger households.

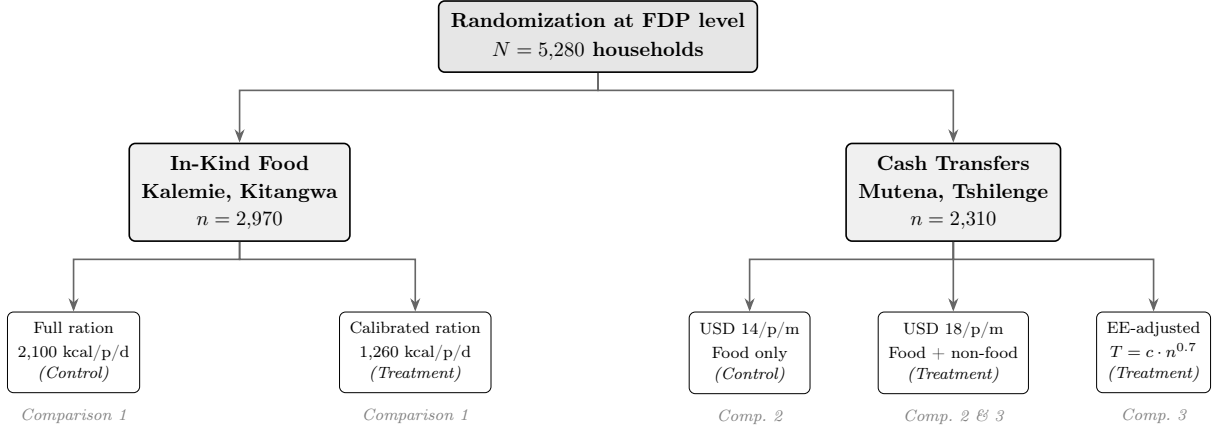
Sources of variation. The design generates three distinct comparisons, each addressing a separate research question:

1. **In-kind ration reduction:** Full ration vs. calibrated ration (40% lower) within in-kind FDPs. This comparison identifies the welfare cost of needs-based calibration.
2. **Cash transfer expansion:** USD 14 (food only) vs. USD 18 (food + non-food) within cash FDPs. This identifies the marginal value of covering non-food needs.
3. **Economies of scale:** EE-adjusted transfer vs. prorated USD 18 within cash FDPs. This tests whether equivalence-scale adjustments achieve comparable welfare outcomes at lower cost.

In addition, the cross-modality comparison of in-kind full rations against USD 14 cash transfers enables a triple difference-in-differences (DiD) analysis of the relative effectiveness of food versus cash. Finally, within each modality, the coexistence of uniform and differentiated FDPs allows us to test whether non-uniform transfers generate community-level tensions, exploiting the FDP-level random assignment.

Figure 1 summarises the experimental structure.

Figure 1: Study Design: Randomization at Food Distribution Point Level



Sample and power. The study is powered to detect a minimum change of 10% in the Food Consumption Score (FCS), the primary outcome. The total planned sample comprises 5,280 households: 2,970 in the food arm and 2,310 in the cash arm, equally distributed across treatment and control FDPs within each modality. Sample sizes were inflated to account for anticipated attrition and non-response.

4 Conceptual Framework

The study addresses four interrelated research questions, each grounded in a distinct but complementary analytical lens. Together, these frameworks provide a unified structure for interpreting the welfare consequences of changes in transfer design across different margins of program variation.

Threshold effects and non-linear welfare responses to needs-based transfer calibration. The first and central question asks what happens to household welfare when transfer values are calibrated to estimated needs rather than set uniformly. The key analytical question is whether welfare outcomes respond proportionally to this calibration or whether non-linearities arise. We frame this through the lens of threshold effects and non-linear marginal returns near subsistence. Standard welfare analysis assumes smooth, concave utility: a transfer calibrated to cover 60% of the gap rather than the full gap should reduce welfare proportionally or less. However, when households operate near

subsistence, critical consumption thresholds may generate disproportionate responses. If the needs-based calibration still leaves some households below such a threshold, the welfare impact will exceed what the proportional benchmark would predict. The existing literature on transfer size draws primarily from development programs where beneficiaries are further from subsistence (Shapiro and Haushofer, 2016; Balboni et al., 2022). In humanitarian settings, where beneficiaries are poorer and coping margins thinner, the welfare response may be substantially larger. This framework predicts that: (i) welfare effects will exceed proportional benchmarks; (ii) households closer to subsistence will experience disproportionately larger effects; and (iii) coping strategies will intensify non-linearly.

Intra-household allocation under ration reductions. The second question examines how different household members—particularly women and children—are affected by reductions in in-kind food rations. We draw on the household production framework (Becker, 1981) and the collective household model (Chiappori, 1992). In-kind food operates as a partially public good within the household, and when the total ration is reduced, the allocation rule determines which members absorb the difference. This framework predicts that: (i) MAD (children) and MDDW (women) will respond differently to the same reduction; and (ii) intra-household allocation norms will partially shield children’s nutritional outcomes at the expense of women’s dietary diversity.

Social comparison and fairness norms under scarcity. The third question asks whether distributing non-uniform transfer values within the same community generates social tensions, and whether this depends on modality. We frame this through models of inequality aversion (Fehr and Schmidt, 1999) and the literature on aid and social cohesion (Croston et al., 2014; Della Guardia et al., 2022). The social cost depends on the *observability* of the transfer: in-kind food is distributed publicly, while cash is received privately. This framework predicts that: (i) non-uniform in-kind transfers will increase community tensions; (ii) the same variation in cash will produce smaller or null effects; and (iii) perceptions of unfairness, rather than absolute deprivation, drive the social costs.

Economies of scale and the optimal transfer function. The fourth question evaluates whether adjusting transfer values for economies of scale can achieve equivalent welfare outcomes at lower cost. This draws on the equivalence-scale literature (Deaton and Paxson, 1998; Lewbel, 1997). Standard humanitarian practice sets $T = c \times n$; an alternative $T = c \times n^\beta$ with $\beta < 1$ would deliver equivalent outcomes while freeing resources. This framework predicts that: (i) no significant welfare differences will emerge between EE-adjusted and prorated transfers; and (ii) the cost savings are quantifiable as additional households covered.

5 Data and Empirical Strategy

5.1 Data

The study draws on two primary data sources: household surveys and focus group discussions. Household-level data come from WFP’s post-distribution monitoring (PDM) surveys,² collected at two points in time: a baseline round conducted before the intervention began and an endline round approximately six months after the program ended. The two rounds constitute repeated cross-sections rather than a panel: different households may be sampled in each wave, though the sampling frame covers the same villages and treatment arms. The surveys provide detailed information on food security, consumption patterns, household expenditures, and individual-level nutritional indicators for women and children. Focus group discussions were held in selected villages to capture community perceptions, experiences with the assistance, and reactions to changes in transfer values.

The total sample size for the in-kind (food) modality consisted of 2,970 households across the Kalemie and Kitangwa sites, equally divided between treatment and control villages. For the cash modality, the total sample comprised 2,310 households across the Mutema and Tshilenge sites, also equally distributed across villages. Table 1 provides descriptive statistics on the sample.

²PDMs are standardized household surveys conducted by WFP after each distribution cycle to monitor food security, consumption patterns, and coping strategies among beneficiaries. They follow a repeated cross-sectional design, surveying a representative sample of beneficiaries in each round.

Table 1: Descriptive Statistics: Baseline Sample

	Food Arm		Cash Arm	
	Mean/N	SD	Mean/N	SD
<i>Panel A: Sample</i>				
Sites	Kalemie, Kitangwa		Mutena, Tshilenge	
Modality	In-kind food		Cash transfers	
Baseline households	2780		2533	
Control group	1391		800	
Treatment group	1361		1733	
Endline households	2745		2437	
<i>Panel B: Baseline characteristics</i>				
Household size	5.76	2.38	5.64	2.51
Age of household head	40.54	13.76	45.12	14.27
Female-headed HH (%)	0.36	0.48	0.27	0.44
Food Consumption Score	27.87	10.09	28.53	11.96
Dietary Diversity (HDDI)	3.59	1.20	4.08	1.10
Reduced Coping Strategy (rCSI)	13.34	10.25	17.74	12.40
Food expenditure (ln)	11.37	0.68	11.30	0.66
Total expenditure (ln)	11.73	0.68	11.67	0.67
Share food insecure	0.88	0.32	0.86	0.34
Observations	2780		2533	

Notes: Panel A reports sample sizes and site characteristics by modality. Panel B reports baseline means and standard deviations for key household characteristics. Food arm covers Kalemie and Kitangwa (in-kind food assistance); cash arm covers Mutena and Tshilenge (cash-based transfers). In the food arm, the control group receives full rations and the treatment group receives calibrated rations. In the cash arm, the control group receives USD 14/p/m (food only) and the treatment group receives either USD 18/p/m prorated (food + non-food) or the EE-adjusted transfer.

5.2 Empirical Strategy

The empirical strategy maps each research question to a specification that exploits the relevant source of variation.

Community-level tensions and social cohesion. Social outcomes—including self-reported tensions and social cohesion—are measured at the village level using endline data only. Because baseline measures of these outcomes are not available, we estimate an intent-to-treat (ITT) specification that exploits the random assignment of transfer designs at the FDP level:

$$Y_i = \alpha + \beta T_i + \mathbf{X}_i' \theta + \varepsilon_i \quad (1)$$

where T_i indicates assignment to non-uniform transfers within the FDP and $\mathbf{X}_i$ includes baseline household covariates. The identification of β relies on the randomization at the FDP level, supported by the baseline balance documented in Tables A1–A2. We estimate this separately for in-kind and cash villages, and test for differential effects across modalities by adding a $T_i \times \text{Cash}_i$ interaction term in a pooled specification.

Household-level welfare effects. To estimate the impact of transfer value changes on household outcomes, we use a difference-in-differences (DiD) design exploiting the repeated cross-sectional structure of the data (baseline and endline rounds):

$$Y_{it} = \alpha + \gamma_g + \delta_t + \beta(T_i \times \text{Post}_t) + \mathbf{X}'_{it}\theta + \varepsilon_{it} \quad (2)$$

where γ_g captures group (treatment arm) fixed effects, δ_t a common time effect, T_i indicates assignment to the treatment arm (calibrated ration in the food modality; USD 14 in the cash modality), and $\mathbf{X}_{it}$ is a vector of household controls. With two time periods and two groups, this is the canonical two-way fixed effects (TWFE) estimator. However, because the data are repeated cross-sections rather than a household panel, we cannot include household-level fixed effects to absorb time-invariant unobservables. Instead, we rely on matching methods to improve covariate balance between treatment and control groups within each period, ensuring that the comparison of group-level changes over time is not confounded by compositional differences across survey rounds.

We estimate three variants: (i) OLS DiD with regression controls; (ii) doubly robust DiD (DR-DiD), which combines matching with outcome regression—our preferred specification, as it provides consistent estimates if either the matching model or the outcome model is correctly specified (Sant’Anna and Zhao, 2020); and (iii) inverse probability weighting (IPW) without regression controls. Throughout, standard errors are clustered at the village level to account for the FDP-level assignment. This specification is applied to both the in-kind arm (full vs. calibrated ration) and the cash arm (USD 14 vs. USD 18), as well as to the economies-of-scale comparison within the cash arm. The coefficient β identifies the causal effect under the assumption that, absent the intervention, treatment

and control groups would have followed the same time trend. Because transfer designs were randomly assigned at the FDP level, treatment and control households within each modality share the same local context—markets, security conditions, seasonal shocks—making common time trends a plausible assumption. The matching methods described above further strengthen this by ensuring that the composition of treatment and control groups is balanced on observable characteristics in each period, so that measured changes over time reflect program impacts rather than compositional shifts.

Modality comparison. To compare cash and in-kind transfers, we estimate a triple difference-in-differences:

$$Y_{it} = \alpha + \beta_1(T_i \times \text{Post}_t) + \beta_2(\text{Cash}_i \times \text{Post}_t) + \beta_3(T_i \times \text{Post}_t \times \text{Cash}_i) + \mathbf{X}'_{it}\theta + \gamma_i + \delta_t + \varepsilon_{it} \quad (3)$$

where β_3 captures the differential effect of the transfer calibration under cash relative to in-kind. Outcomes are standardized to z-scores to enable comparison across indicators with different units.

5.3 Baseline Balance

To ensure experimental integrity, we present balance tests for the different transfer variation arms. The balance test compares baseline characteristics across treatment groups to verify that randomization achieved comparable distributions of key household attributes, such as demographic composition, baseline food security status, and expenditure levels. Because treatment assignment was randomized at the FDP level, households within the same modality are drawn from the same operational environment, lending credibility to the common trends assumption required by the DiD design. The matching estimators further reinforce this by balancing observable characteristics between treatment and control groups within each survey round, so that before-and-after comparisons capture program effects rather than compositional differences across rounds. All balance tests are reported in the Appendix: Tables [A1](#) and [A2](#) cover the within-modality arms, and Table [A3](#) provides a balance comparison across modalities (in-kind full ration vs. cash

USD 14).

6 Results

We begin by examining whether non-uniform transfer values generate community tensions and affect social cohesion, exploiting the FDP-level randomization directly. We then present the household-level effects of in-kind ration reductions, examine how these reductions are transmitted within the household to women and children, and analyze heterogeneities by baseline vulnerability, household size, and gender of the household head. Next, we turn to the cash arm and evaluate whether economies-of-scale adjustments can achieve comparable outcomes at lower cost. We then compare cash versus in-kind transfers using a triple difference-in-differences design and present program design simulations that quantify the cost–welfare trade-offs of alternative transfer configurations. Finally, we report qualitative findings from focus group discussions.

6.1 Social Cohesion and Community Tensions

Table 2 reports the effects of uniform versus non-uniform transfer values on reported community tensions, disaggregated by modality (in-kind food vs. cash). The analysis focuses on four outcomes related to conflict and perceived fairness within the village.

In villages receiving in-kind food, variation in transfer values significantly increased the probability that community members perceived WFP aid as increasing conflict, with an estimated effect size of 0.28 percentage points. This suggests that when in-kind assistance is distributed unequally across households within the same village, it can generate perceptions of unfairness that contribute to social tension. Other indicators, such as reports of excluded groups needing assistance or inter-community problems, show small and statistically insignificant changes.

In contrast, no statistically significant effects are observed in cash villages across any of the four outcomes. Even though the point estimates are sometimes positive, the confidence intervals include zero, and the estimates are considerably smaller than in the

in-kind setting. This pattern supports the interpretation that cash transfers are less visible and therefore less likely to provoke social comparison or conflict when distributed unevenly.

Overall, the results suggest that non-uniform transfer values can create community tensions in in-kind settings, but cash transfers mitigate this risk, possibly due to their lower visibility and more private nature. Consistent with this interpretation, the Social Cohesion Index (Anderson, 2008) for the cash arm shows no significant change (0.06, n.s.; Panel B of Table 2), indicating that non-uniform cash transfers do not erode broader social cohesion. The individual components of the index are reported in Table A7 in the Appendix.

Table 2: Effects of Non-Uniform Transfer Values on Community Tensions and Social Cohesion

	(1)	(2)	(3)	(4)
	In-Kind		Cash	
	Coeff.	Ctrl Mean	Coeff.	Ctrl Mean
<i>Panel A: Community tensions</i>				
Experienced conflict/tensions in last 30 days	-0.00 (0.00)	0.01	0.00 (0.01)	0.01
Reports of excluded groups needing assistance	0.00 (0.02)	0.17	0.02 (0.04)	0.19
Perceived that WFP aid increased conflict	0.14* (0.07)	0.41	0.06 (0.06)	0.54
Distribution caused inter-community problems (binary)	0.01 (0.01)	0.03	-0.00 (0.02)	0.03
<i>Panel B: Social cohesion (cash arm)</i>				
Social Cohesion Index (Anderson 2008)			0.06 (0.12)	-0.04
Controls	Yes		Yes	
Observations	2701		1694	

Notes: Panel A: Columns (1)–(2) report ITT estimates for villages receiving in-kind food, with control group means. Columns (3)–(4) report the same for cash villages. Panel B: Social Cohesion Index constructed using the Anderson (2008) inverse-covariance-weighted approach for the cash arm only; individual components reported in the Appendix. All specifications control for baseline covariates. Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Impact of In-Kind Transfer Reductions

Table 3 reports the impact of a 40% reduction in in-kind food rations on household welfare. We present estimates from three specifications: OLS difference-in-differences, doubly robust DiD (DR-DiD), and inverse probability weighting (IPW). Throughout, we focus

the discussion on the DR-DiD estimates (column 2), which combine regression adjustment with propensity score weighting and serve as our preferred specification. Results are broadly consistent across methods.

Food security and expenditure. The reduction in rations generates large and statistically significant declines across all food security indicators. The Food Consumption Score falls by 18.14 points ($p < 0.01$), equivalent to approximately 7% of the control group mean, while the Household Dietary Diversity Index declines by 2.00 points ($p < 0.01$), a 44% reduction relative to baseline. The share of food-insecure households increases by 35 percentage points ($p < 0.01$), indicating a substantial shift in the distribution of food security status; Table A5 (Appendix) decomposes this shift across the poor, borderline, and acceptable FCS categories. These effects are robust to reweighting: the IPW estimates in column 3 are somewhat larger in magnitude but qualitatively consistent.

Household expenditure contracts across all categories. Log food expenditure falls by 0.38 log points ($p < 0.01$), corresponding to an approximate 31% decline. Non-food expenditure declines by 0.44 log points ($p < 0.01$), or roughly 36%, and total expenditure falls by 0.39 log points ($p < 0.01$). That non-food expenditure contracts as sharply as food expenditure indicates that households were unable to smooth consumption by reallocating spending across categories; instead, the reduction in food assistance propagated broadly across the household budget.

Expenditure decomposition. Table A4 (Appendix) decomposes these expenditure effects by food group and non-food category. The reductions are strikingly concentrated in food items directly substitutable for the in-kind ration. Expenditure on cereals falls by 1.71 log points ($p < 0.01$, an 82% decline), pulses by 2.64 log points ($p < 0.01$, 93%), and oils and fats by 3.15 log points ($p < 0.01$, 96%). The most severe contractions appear in higher-value food groups: meat and fish expenditure declines by 4.53 log points ($p < 0.01$), representing a near-complete collapse (99%), and fruit expenditure falls by 2.77 log points ($p < 0.01$, 94%). In contrast, expenditure on tubers—a locally produced staple less dependent on market purchase—is unchanged, and vegetable expenditure shows no

significant decline. This pattern suggests that households prioritize calorie-dense local staples at the expense of dietary diversity and nutritionally rich foods when assistance is reduced.

Among non-food categories, education expenditure falls sharply by 1.12 log points ($p < 0.01$, 67%), savings decline by 0.76 log points ($p < 0.01$, 53%), and debt repayment decreases by 0.74 log points ($p < 0.10$, 52%). Health expenditure, by contrast, does not change significantly. The contraction in education spending and savings points to potentially lasting effects that extend beyond immediate food security, as households disinvest from human capital and deplete already limited buffers.

Coping strategies. The aggregate reduced Coping Strategy Index (rCSI) does not change significantly (Panel B of Table 3), but this masks meaningful heterogeneity across individual coping mechanisms. Borrowing food or relying on help from friends and relatives increases by 0.39 days per week ($p < 0.01$), consistent with households activating social networks to compensate for reduced assistance. Other coping dimensions—consuming less preferred foods, reducing the number of meals, reducing portion sizes, and restricting adult intake for children—show no statistically significant changes in the food arm, suggesting that the primary margin of adjustment is increased reliance on informal transfers rather than further dietary compression.

Table 3: Impact of Needs-Based Transfer Calibration: In-Kind Food Arm

	(1) OLS	(2) DR-DiD	(3) IPW
<i>Panel A: Food security and expenditure</i>			
Food Consumption Score	-18.06*** (0.77)	-18.14*** (0.77)	-18.98*** (0.91)
Household Dietary Diversity	-1.98*** (0.10)	-2.00*** (0.10)	-2.13*** (0.13)
Reduced Coping Strategy	0.75 (0.84)	0.87 (0.83)	1.13 (0.84)
Food Expenditure - ln	-0.37*** (0.04)	-0.38*** (0.04)	-0.45*** (0.05)
Non Food Expenditure - ln	-0.44*** (0.07)	-0.44*** (0.07)	-0.58*** (0.07)
Total Expenditure - ln	-0.39*** (0.04)	-0.39*** (0.04)	-0.48*** (0.04)
Share of Food Insecure HHs	0.34*** (0.03)	0.35*** (0.03)	0.37 (0.00)
<i>Panel B: Coping strategy components</i>			
Consume less preferred or cheaper foods	0.09 (0.24)	0.11 (0.24)	0.15 (0.23)
Borrow food or rely on help from friends/rela...	0.38** (0.14)	0.39*** (0.15)	0.49*** (0.16)
Reduce number of meals per day	0.15 (0.19)	0.17 (0.19)	0.22 (0.19)
Reduce portion sizes at meals	-0.10 (0.19)	-0.08 (0.18)	-0.07 (0.19)
Restrict adult intake for children	-0.05 (0.12)	-0.03 (0.12)	-0.06 (0.12)
Controls	Yes	Yes	No
Weights	None	DR	IPW
Observations	5012	5012	5012

Notes: Each row reports the DiD coefficient on the interaction of treatment receipt and post-intervention period. Column (1) reports OLS estimates. Column (2) uses doubly robust weights (inverse probability weighting combined with outcome regression). Column (3) uses inverse probability weights without controls. Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Intra-Household Impacts: Women and Children

A central question in the design of humanitarian assistance is whether changes in transfer values are transmitted uniformly within the household or whether they disproportionately affect more vulnerable members. We examine this by comparing nutritional outcomes for women and children in the food arm, where the 40% ration reduction provides a sharp

test of intra-household allocation under resource contraction.

Effects of in-kind ration reductions. Panel C of Table 3 and Table 4 report effects on individual-level nutritional outcomes in the food arm. The Minimum Dietary Diversity for Women (MDDW) declines by 0.61 points ($p < 0.01$), with the probability of consuming five or more food groups falling by 8 percentage points ($p < 0.01$). These magnitudes indicate that the ration reduction disproportionately erodes dietary quality for women. The Minimum Acceptable Diet for children (MAD) shows a small negative point estimate (-0.02), which is marginally significant in some specifications but not in the DR-DiD. Minimum meal frequency for children is unaffected. This asymmetry suggests that intra-household allocation norms partially shield children’s feeding practices at the expense of women’s dietary diversity: when resources contract, women appear to absorb a larger share of the nutritional cost.

Table 4: Intra-Household Effects of In-Kind Ration Reduction: Women and Children

	(1) OLS	(2) DR-DiD	(3) IPW
<i>Panel A: Women’s outcomes</i>			
Minimum Dietary Diversity (Women)	-0.59*** (0.14)	-0.61*** (0.14)	-0.53*** (0.15)
Dietary Diversity (5+ food groups)	-0.08*** (0.01)	-0.08*** (0.01)	-0.08*** (0.02)
<i>Panel B: Children’s outcomes</i>			
Minimum Acceptable Diet (Children)	-0.02* (0.01)	-0.02 (0.01)	-0.02* (0.01)
Minimum Meal Frequency (children)	0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)
Controls	Yes	Yes	No
Weights	None	DR	IPW
Observations	5012	5012	5012

Notes: Panel A reports effects on women’s dietary diversity (MDDW and 5+ food groups). Panel B reports effects on children’s nutrition (MAD, MMF). Same specifications as Table 3. Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Heterogeneities

We examine whether the effects of the ration reduction vary along three dimensions: baseline vulnerability, household size, and gender of the household head. Table A6 (Appendix) reports DR-DiD estimates separately for each subgroup. The reference category throughout is the most vulnerable group, which continued to receive full rations and therefore shows no treatment effect by construction.

Baseline vulnerability. The negative effects of the ration cut are present across both the medium and least vulnerable terciles, but their relative magnitude reveals an important pattern. For the FCS, the decline is larger among the least vulnerable households (-17.25 , $p < 0.01$) than among medium-vulnerability households (-15.37 , $p < 0.01$). A similar ordering holds for total expenditure (-0.41 vs. -0.36 log points, both $p < 0.01$) and the share of food-insecure households (0.31 vs. 0.24 percentage points, both $p < 0.01$). This pattern is consistent with the hypothesis that households classified as least vulnerable at baseline had higher initial consumption levels that were partly sustained by the full ration; once the ration was cut, the absolute contraction in consumption was at least as large for this group. In contrast, the MDDW decline is concentrated among medium-vulnerability households (-0.63 , $p < 0.01$) relative to the least vulnerable (-0.39 , $p < 0.05$), suggesting that women’s dietary quality is more sensitive to the reduction in households that were already closer to the margin of adequate nutrition.

Household size. The effects are broadly comparable across household size categories, with FCS declines of 17.23 points for small households (1–4 members), 18.54 for medium households (5–7), and 17.93 for large households (8+), all significant at the 1% level. Medium-sized households show the largest point estimate for the FCS and the highest increase in the share of food-insecure members (0.37 , $p < 0.01$), closely matched by large households (0.37 , $p < 0.01$). The MDDW decline is also largest for medium-sized households (-0.64 , $p < 0.01$). The MAD for children shows a marginally significant negative effect only among large households (-0.03 , $p < 0.10$), consistent with greater intra-household dilution of remaining food resources. Overall, the absence of strong

household-size gradients suggests that the ration reduction affects households across the size distribution, with limited scope for larger households to buffer the shock through economies of scale.

Gender of household head. The impacts are similar for female-headed and male-headed households, with FCS declines of 17.79 and 18.25 points, respectively, and comparable magnitudes across all other outcomes. Female-headed households show a marginally smaller decline in food expenditure (-0.32 vs. -0.41 log points) and total expenditure (-0.34 vs. -0.42 log points), though both estimates remain highly significant. The MDDW contraction is nearly identical across groups (-0.58 and -0.59). The rCSI shows a marginally significant increase among male-headed households (1.55 , $p < 0.10$) but not among female-headed households (-0.54 , n.s.), which may reflect differences in coping strategy repertoires or reporting. In sum, the ration reduction does not appear to impose systematically different welfare costs by the gender of the household head, suggesting that the reduction affects households uniformly along this margin.

6.5 Impact of Cash Transfer Variation

We now turn to the cash arm, where the main source of variation is the inclusion of a non-food component: households assigned to the USD 18 transfer received an additional USD 4 per person per month to cover non-food expenditures, while those assigned to the USD 14 transfer received support for food needs only. Table 5 reports the DiD estimates, where the treatment group comprises households receiving USD 14 and the comparison group those receiving USD 18.

Food security and expenditure. The results indicate that households receiving the lower transfer value experience significant welfare losses relative to those receiving the expanded transfer. The FCS declines by 13.09 points in the DR-DiD specification ($p < 0.01$), while the HDDI falls by 0.88 points ($p < 0.01$). These magnitudes are substantial, though somewhat smaller than those observed in the food arm, consistent with the fact that the differential between the two cash transfer values (USD 4 per person) represents a

smaller proportional change in total support than the 40% ration cut. The rCSI increases by 3.73 points ($p < 0.01$), indicating a meaningful deterioration in coping behavior—a result that contrasts with the food arm, where the aggregate coping index did not change significantly.

Log food expenditure falls by 0.23 log points ($p < 0.01$), non-food expenditure by 0.36 log points ($p < 0.01$), and total expenditure by 0.26 log points ($p < 0.01$). The share of food-insecure households increases by 35 percentage points ($p < 0.01$). The sharper decline in non-food expenditure is consistent with the design of the variation: since the USD 18 transfer explicitly includes a non-food component, its removal forces households to compress spending on non-food items or reallocate from food budgets to cover essential non-food needs.

Coping strategies. Unlike the food arm, the cash arm shows significant increases across multiple coping dimensions. Borrowing food or relying on help increases by 0.43 days per week ($p < 0.01$), reducing portion sizes increases by 0.70 days ($p < 0.05$), and restricting adult intake for children rises by 0.49 days ($p < 0.01$). The activation of these more severe coping mechanisms—particularly restricting adult food consumption to protect children—suggests that the removal of the non-food component places households under more acute pressure than the in-kind ration cut, possibly because cash recipients have fewer alternative sources of support.

Table 5: Impact of Needs-Based Transfer Calibration: Cash Arm

	(1) OLS	(2) DR-DiD	(3) IPW
<i>Panel A: Food security and expenditure</i>			
Food Consumption Score	-13.18*** (1.63)	-13.09*** (1.63)	-13.39*** (1.66)
Household Dietary Diversity	-0.87*** (0.13)	-0.88*** (0.12)	-0.90*** (0.12)
Reduced Coping Strategy	3.73*** (1.32)	3.73*** (1.32)	4.15*** (1.37)
Food Expenditure - ln	-0.24*** (0.06)	-0.23*** (0.06)	-0.24*** (0.06)
Non Food Expenditure - ln	-0.37*** (0.10)	-0.36*** (0.10)	-0.39*** (0.11)
Total Expenditure - ln	-0.27*** (0.06)	-0.26*** (0.06)	-0.28*** (0.06)
Share of Food Insecure HHs	0.35*** (0.06)	0.35*** (0.06)	0.36 (0.00)
<i>Panel B: Coping strategy components</i>			
Consume less preferred or cheaper foods	0.35 (0.38)	0.38 (0.38)	0.46 (0.40)
Borrow food or rely on help from friends/rela...	0.42*** (0.15)	0.43*** (0.15)	0.47*** (0.15)
Reduce number of meals per day	0.33 (0.30)	0.33 (0.30)	0.39 (0.31)
Reduce portion sizes at meals	0.67** (0.31)	0.70** (0.31)	0.76** (0.31)
Restrict adult intake for children	0.51*** (0.15)	0.49*** (0.15)	0.53*** (0.16)
Controls	Yes	Yes	No
Weights	None	DR	IPW
Observations	4227	4227	4227

Notes: Same specifications as Table 3. Treatment is receipt of the USD 14 transfer (food component only) relative to USD 18 (food + non-food). Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.6 Economies of Scale

Table 6 reports the impact of using economies of scale (EE) in setting transfer values relative to the prorated approach, disaggregated by household size: small (1–4 members), medium (5–7), and large (8+). Across all groups, the results suggest that applying EE instead of a flat transfer value of USD 18 per person per month yields no significant negative effects on household welfare outcomes.

Among small households, no substantial differences are detected between the EE and prorated approaches. Similarly, for medium-sized and large households, indicators such as FCS, HDDI, food and non-food expenditures, and total consumption remain stable across both approaches. The rCSI shows a marginally significant increase overall (2.83, $p < 0.10$) and for medium households (3.66, $p < 0.05$), suggesting some adjustment in coping strategies, but this is not accompanied by significant deterioration in food security or expenditure. Slight variations in individual outcomes are observed, but none indicate systematic disadvantages under the EE design.

Importantly, the results suggest that using EE could generate cost savings without compromising key outcomes. These savings could be reinvested to expand program coverage, allowing assistance to reach a larger number of vulnerable households without reducing the quality of support provided to those already enrolled.

Table 6: Economies of Scale vs. Prorated Transfer Values by Household Size

	(1) Overall	(2) Small HH (1–4)	(3) Medium HH (5–7)	(4) Large HH (8+)
Food Consumption Score	-0.59 (1.92)	-0.74 (1.89)	-0.61 (2.02)	1.13 (2.43)
Household Dietary Diversity	0.03 (0.16)	0.02 (0.18)	0.05 (0.17)	0.08 (0.21)
Reduced Coping Strategy	2.83* (1.67)	1.92 (1.75)	3.66** (1.82)	2.21 (2.01)
Food Expenditure - ln	0.12 (0.08)	0.10 (0.09)	0.15* (0.08)	0.09 (0.08)
Non Food Expenditure - ln	-0.06 (0.13)	-0.09 (0.13)	0.03 (0.14)	-0.14 (0.16)
Total Expenditure - ln	0.09 (0.08)	0.08 (0.09)	0.14 (0.09)	0.04 (0.10)
Expenditure on health - ln	-0.49 (0.30)	-0.50 (0.35)	-0.51 (0.34)	-0.24 (0.38)
Expenditure on education - ln	-0.77 (0.49)	-0.63 (0.66)	-1.08** (0.54)	-0.60 (0.45)
Minimum Acceptable Diet (Children)	0.03 (0.02)	-0.01 (0.04)	0.04 (0.03)	0.03 (0.02)
Minimum Dietary Diversity (Women)	0.15 (0.22)	0.03 (0.28)	0.18 (0.24)	0.24 (0.27)
Controls	Yes		Yes	
Observations	2473		2473	

Notes: Column (1) reports the overall DiD effect of the economies-of-scale (EE) transfer design relative to prorated transfers. Columns (2)–(4) interact the treatment with household size categories to test for differential effects. All specifications include baseline controls and cluster standard errors at the village level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.7 Cash vs. In-Kind

Table 7 presents the comparative effects of cash versus in-kind transfers on household welfare outcomes using a triple difference-in-differences design. All outcomes are standardized to z-scores to enable comparison across indicators with different scales.

The results indicate that cash transfers significantly outperform in-kind assistance across several dimensions. The FCS is 0.45 SD higher under cash ($p < 0.01$), and the HDDI is 0.96 SD higher ($p < 0.01$), showing large positive effects on food security and dietary variety. Food expenditure is 0.25 SD higher ($p < 0.05$) and total expenditure 0.22 SD higher ($p < 0.05$), suggesting that cash enables households to allocate resources

more flexibly across essential needs. The rCSI is 0.28 SD higher under cash ($p < 0.05$), indicating less reliance on negative coping strategies.

There are no significant impacts on non-food, health, or education expenditure, highlighting that the main benefits of cash appear concentrated in food-related outcomes and flexible household spending. Nutritional indicators for women (MDDW) and children (MAD) show no significant differential effects across modalities, suggesting that the advantage of cash operates primarily through household-level consumption rather than individual dietary composition. Overall, the evidence shows that cash assistance enabled households to expand consumption, improve dietary diversity, and better meet basic needs compared to in-kind food support.

Table 7: Triple Difference-in-Differences: Cash vs. In-Kind Transfers

	(1) OLS	(2) DR-DiD	(3) IPW
Food Consumption Score (z)	0.43*** (0.15)	0.45*** (0.15)	0.48*** (0.16)
Household Dietary Diversity (z)	0.93*** (0.13)	0.96*** (0.13)	1.00*** (0.14)
Reduced Coping Strategy (z)	0.30** (0.14)	0.28** (0.14)	0.27* (0.14)
Food Expenditure - ln (z)	0.22** (0.11)	0.25** (0.11)	0.34*** (0.12)
Non Food Expenditure - ln (z)	0.09 (0.11)	0.10 (0.11)	0.18 (0.11)
Total Expenditure - ln (z)	0.20* (0.10)	0.22** (0.10)	0.31*** (0.12)
Expenditure on health - ln (z)	-0.08 (0.10)	-0.06 (0.11)	-0.03 (0.11)
Expenditure on education - ln (z)	0.14 (0.10)	0.14 (0.10)	0.26** (0.12)
Minimum Acceptable Diet (Children) (z)	0.05 (0.18)	0.06 (0.18)	0.08 (0.18)
Minimum Dietary Diversity (Women) (z)	-0.11 (0.14)	-0.12 (0.14)	-0.13 (0.14)
Controls	Yes	Yes	No
Weights	None	DR	IPW

Notes: Each row reports the triple-interaction coefficient (Post $\times$ Treatment $\times$ Cash), which captures the differential effect of the transfer calibration for cash relative to in-kind modality. All outcomes are standardized to z-scores (mean 0, SD 1) using the control group distribution. Column (1) is OLS. Column (2) is doubly robust. Column (3) uses inverse probability weights. Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.8 Program Design Simulations

Economies of scale. Table 8 compares the total program cost under three transfer designs: the prorated USD 14 (food only), the prorated USD 18 (food + non-food), and the economies-of-scale (EE) formula with $\beta = 0.7$. The EE approach generates savings of approximately USD 30,690 relative to the USD 18 design, sufficient to extend coverage to 53 additional households—a 2.2% expansion. A sensitivity analysis varying the equivalence-scale parameter from $\beta = 0.6$ to $\beta = 0.9$ shows that even conservative assumptions about economies of scale yield positive savings, ranging from 0.9 to 2.5% expansion in coverage. Given the null effects of the EE design on household welfare documented in Table 6, these results suggest that accounting for economies of scale offers a feasible margin for efficiency gains without compromising program effectiveness.

Needs-based ration calibration. The food ration reduction offers a second margin for cost optimization. Under the full ration design, each beneficiary receives a food basket valued at approximately USD 14 per person per month. The calibrated ration, set at 60% of the full basket, reduces the per-person cost to approximately USD 8.40 per month—a saving of USD 5.60 per person per month. For a household of average size (approximately 5.7 members) over the six-month program cycle, this implies a saving of approximately USD 192 per household. Applied to the roughly 1,485 households that received the calibrated ration, the aggregate savings amount to approximately USD 285,000, which could finance full-ration coverage for an additional 625 households.

However, these savings must be weighed against the welfare costs documented in Section 6.2, which are disproportionately large relative to the size of the transfer cut. A 40% reduction in rations produces a 44% decline in dietary diversity, a 31% decline in food expenditure, and near-complete collapses in spending on nutrient-rich food groups. Education expenditure falls by 67% and savings by 53%, indicating effects that extend well beyond immediate food security. The aggregate savings of USD 285,000 would need to offset the welfare deterioration experienced by 1,485 households across multiple dimensions—food security, dietary diversity, expenditure, women’s nutrition, and

human capital investment. The magnitude and breadth of these welfare losses suggest that the needs-based ration calibration, while generating fiscal savings, imposes costs that substantially exceed the value of the resources freed. This contrasts sharply with the economies-of-scale and modality-switch margins, where efficiency gains come at no measurable welfare cost or with positive welfare returns.

Table 8: Cost Simulation: Economies of Scale vs. Prorated Transfer Values

	Prorated (USD 14)	Prorated (USD 18)	EE ($\beta = 0.7$)
Mean cost per HH (USD)	457.05	587.64	575.05
Total program cost (USD)	1,113,840	1,432,080	1,401,390
Total savings vs. USD 18	–	–	30,690
Additional HH covered	–	–	53
Expansion (%)	–	–	2.2%
<i>Sensitivity: Total savings by equivalence-scale parameter β</i>			
$\beta = 0.6$	–	–	35,610 (2.5)
$\beta = 0.7$	–	–	30,690 (2.2)
$\beta = 0.8$	–	–	23,064 (1.6)
$\beta = 0.9$	–	–	12,814 (0.9)

Notes: The prorated approach sets the transfer at a fixed per-capita rate multiplied by household size. The EE approach uses an equivalence-scale formula $T = c \times n^\beta$, with $\beta = 0.7$ as the baseline and c calibrated so that average program cost matches the prorated USD 18 scenario. “Additional HH covered” is the number of households that could be added to the program using the savings, valued at the average EE cost per household. The sensitivity panel varies β from 0.6 to 0.9.

Simulation 1: Modality switch. The first simulation compares the cost of delivering assistance via in-kind food versus cash. According to WFP’s country-level evaluation (OEV/2023/013), the direct cost per beneficiary was USD 48 for in-kind and USD 15 for cash-based transfers in 2023. This 3.2:1 ratio reflects the transport, warehousing, and logistics overhead inherent in commodity-based delivery. Applying this ratio to the study’s per-household transfer cost, the in-kind full ration (valued at USD 14 per person per month) costs approximately USD 1,463 per household over 6 months. The cash transfer covering food and non-food needs (USD 18 per person per month, prorated) costs USD 588 per household—less than half the in-kind cost, while also covering non-food expenditures. The triple DiD estimates show that cash delivers superior welfare outcomes (+0.45 SD FCS, +0.96 SD HDDI), so the modality switch both expands coverage and improves outcomes per household reached.

Simulation 2: Economies of scale within cash. The second simulation operates within the cash modality, comparing the prorated USD 18 approach (USD 588 per household over 6 months) with the EE formula ($\beta = 0.7$, USD 575 per household)—a saving of USD 13 per household, or 2.2% of program cost. While the absolute savings are modest compared to the modality switch, they come at no measurable welfare cost (Section 6.6) and can be implemented without changing delivery infrastructure.

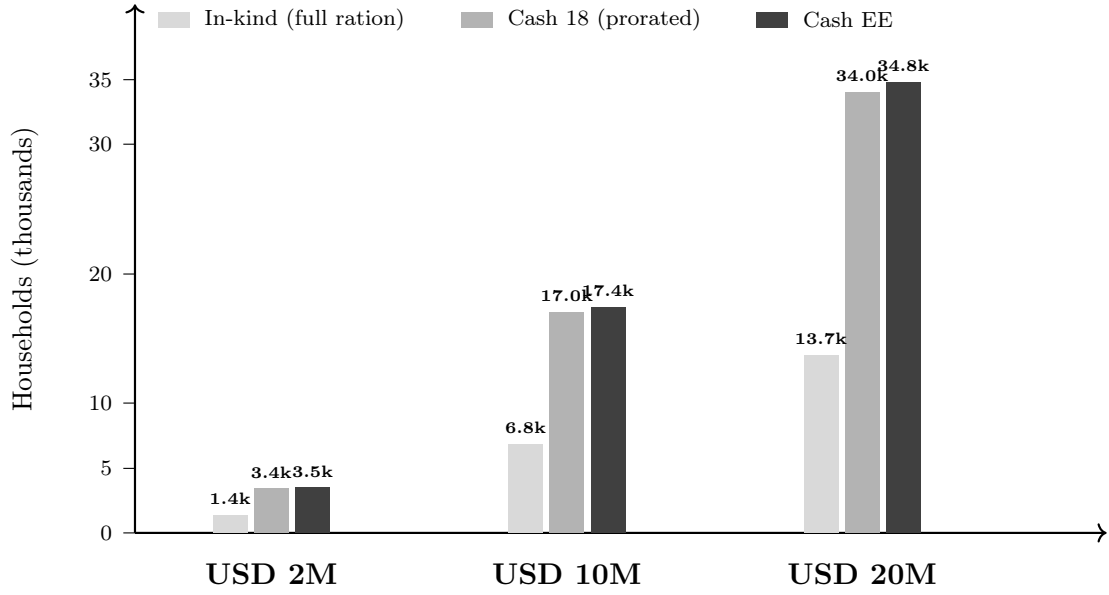
Combined simulation. Table 9 and Figure 2 present the coverage implications of both design margins across three illustrative budget levels. Panel A shows the baseline: in-kind delivery of the full food ration. Panels B and C show the two cash alternatives: prorated USD 18 (food + non-food) and EE-adjusted USD 18. At every budget level, cash covers approximately 2.5 times as many households as in-kind, even when delivering a more comprehensive transfer (food + non-food vs. food only). The EE adjustment adds a further 2.2% coverage gain on top of the modality switch. These gains are not offset by welfare losses: cash outperforms in-kind on all food security measures, and EE produces equivalent outcomes to prorated transfers across all household sizes.

Table 9: Program Design Simulations: Coverage at Constant Budget

	USD 2M	USD 10M	USD 20M
<i>Panel A: In-kind, full ration (USD 14 food value)</i>			
Cost per HH (6 months)		USD 1,463	
Households covered	1,367	6,836	13,671
<i>Panel B: Cash 18, food + non-food, prorated</i>			
Cost per HH (6 months)		USD 588	
Households covered	3,401	17,007	34,014
Additional HH vs. in-kind	+2,034	+10,171	+20,343
Expansion vs. in-kind	+149%	+149%	+149%
<i>Panel C: Cash EE, food + non-food, EE-adjusted</i>			
Cost per HH (6 months)		USD 575	
Households covered	3,478	17,391	34,783
Additional HH vs. in-kind	+2,111	+10,555	+21,112
Expansion vs. in-kind	+154%	+154%	+154%
Additional HH vs. Cash 18	+77	+384	+769

Notes: Panel A: in-kind per-HH cost derived by applying the 3.2:1 cost ratio from the WFP evaluation (OEV/2023/013; USD 48 vs. USD 15 per beneficiary) to the study's cash transfer cost. Panels B–C: cash at USD 18/p/m with prorated or EE allocation, per-HH costs from Table 8. Average HH size = 5.7 members. Cash 18 covers food + non-food needs (USD 4/p/m top-up). Welfare effects: cash yields +0.45 SD FCS vs. in-kind (Table 7); EE shows no welfare differences vs. prorated (Table 6).

Figure 2: Household Coverage by Modality and Transfer Design at Constant Budget



Notes: Three bars per budget level show households covered under: (i) in-kind full ration at USD 14 food value (USD 1,463/HH total cost); (ii) Cash 18: prorated USD 18/p/m covering food + non-food (USD 588/HH); (iii) Cash EE: USD 18/p/m with economies-of-scale adjustment (USD 575/HH). Both cash designs cover food and non-food needs while reaching 2.5× as many households as in-kind. Per-HH costs from Table 8; in-kind cost ratio from WFP evaluation (OEV/2023/013).

6.9 Qualitative Findings

Community Perceptions of Ration Differentiation. Strong resistance emerged across focus group discussions—particularly among women’s groups—regarding the use of different ration sizes for households with varying vulnerability classifications. A commonly voiced sentiment was: “We all have problems and suffer, so assist everyone with the same rations according to the size of the household.” This perspective, echoed in nearly all focus groups, reflects a shared sense of injustice and confusion over the targeting criteria. Many participants described the differentiated ration system as “discriminatory,” pointing out that households classified as “very vulnerable” were often just as, if not more, in need than those labeled “extremely vulnerable.”

The approach also fueled perceptions of favoritism and sowed resentment within communities. In one village inhabited by Twa populations, some participants even interpreted the reduced rations as being linked to ethnic discrimination, despite efforts to raise awareness and explain the program’s design.

Frustration with Insufficient Rations and Social Tensions. Dissatisfaction with the quantity of food provided under the reduced ration was a consistent theme across focus groups. Many participants, especially women, voiced frustration that the smaller rations were insufficient to meet basic needs, with cereals being singled out as particularly inadequate. The shortfall in essential staples heightened the sense of inequality and deepened frustration among recipients.

The distribution of different ration sizes also introduced social tensions within communities. Women in all focus groups described how the approach created jealousy and frustration, both among households receiving reduced rations and from non-beneficiaries observing the distribution. This sense of unequal treatment fostered stigmatization of those who received the reduced ration, with some women noting that these households were singled out or looked down upon by others in the community. As one group put it, the unequal rations have “caused tension between those receiving reduced and full rations,” underscoring the social risks introduced by the targeting approach.

Feedback Channels and Community Preferences. While Help and Feedback Desks were actively used by communities, they primarily served to address operational issues—such as distribution logistics or registration problems—rather than concerns related to program design. However, complaints related to changes in ration size were frequently received through the *Ligne Verte* (hotline), indicating that dissatisfaction with the new rationing approach was widespread. Interestingly, communities preferred to raise concerns face-to-face, often during program consultations or focus group discussions, rather than through more formal channels. This underscores the importance of maintaining direct, community-level engagement to surface concerns about program fairness and design.

Several examples from the hotline highlight the nature of these complaints. One caller explained: “*Il se plaint de quantité des vivres reçue lors de la première distribution dans son village. Il estime qu’elle est insuffisante.*” Another reported: “*Il se plaint du fait que la quantité des vivres reçue lors des distributions effectuées dans leur village étaient insuffisante par rapport au nombre des personnes de sa famille et il sollicite ainsi une augmentation pour le prochain projet si il a lieu.*” These statements reflect a shared per-

ception that rations did not adequately account for household size and a strong demand for more equitable or tailored distribution in future interventions.

Contextual Challenges in Eastern DRC. Participants and field teams raised serious concerns about the feasibility of implementing a differentiated ration approach in eastern provinces such as North Kivu, South Kivu, and Ituri. In these areas, inter-ethnic tensions are already heightened, and vulnerability often correlates closely with ethnic identity. As a result, any assistance perceived as unequal risks reinforcing or being interpreted through ethnic lenses, potentially exacerbating existing divisions. The risk of aid being seen as favoring one group over another was consistently flagged as a major barrier to this approach in conflict-affected, ethnically diverse areas.

Moreover, the prioritization of internally displaced persons (IDPs) under the current targeting model has already generated tensions between IDPs and host communities, particularly where both groups face high levels of vulnerability. In settings with ongoing conflict and constrained humanitarian access, implementing such an approach without triggering new forms of tension would be highly challenging. Participants emphasized that this would require a robust contextual analysis, including a thorough conflict sensitivity assessment, before rollout could be considered appropriate or safe.

Operational Feasibility. A practical challenge in implementing differentiated transfer values is the operational complexity of determining, communicating, and distributing varying amounts within the same community. In-kind food distributions require physically separating commodities into different ration sizes at distribution points, which increases logistical burden, distribution time, and the potential for errors (Gentilini, 2016). For the in-kind arm of this study, WFP addressed this by issuing color-coded ration cards to distinguish vulnerability groups and calling up beneficiaries by card color and household size. Field teams reported that the approach was logistically feasible but required careful planning between program and supply chain units, additional days of sensitization in each village, and the presence of WFP staff (not only partners) at initial distributions to address beneficiary concerns that reduced rations were the result of diversion.

Cash-based transfers are inherently simpler to differentiate, since the transfer amount can be varied electronically without affecting distribution logistics. The EE-adjusted transfers, which assign a different amount to each household based on size, were implemented through mobile money platforms without additional operational cost. This asymmetry in operational complexity between modalities is itself a relevant finding: the in-kind modality not only costs more to deliver (Section 6.8) and generates more community tensions (Section 6.1), but also faces greater logistical barriers to implementing differentiated transfer designs.

7 Conclusions

This paper exploits three sources of exogenous variation in humanitarian transfer values in the Democratic Republic of the Congo to study how changes in the level, structure, and modality of assistance affect household welfare, intra-household allocation, and community cohesion. The setting—one of the world’s largest and most protracted food crises—makes these questions both urgent and policy-relevant, particularly as declining aid budgets force agencies to choose between reducing transfer values and removing beneficiaries from program rolls.

Our first finding is that needs-based calibration of in-kind rations imposes welfare costs that exceed what proportional benchmarks would predict. A 40% reduction in the food ration leads to an 18-point decline in FCS, a 44% decline in dietary diversity, and a 35 percentage point increase in the share of food-insecure households. Expenditure contracts broadly rather than selectively: food, non-food, education, and savings all decline, with near-complete collapses in spending on nutrient-rich food groups such as meat and fish, fruits, and oils. Households do not smooth consumption across categories; instead, the reduction propagates through the entire budget, suggesting that many households operate near consumption thresholds below which welfare deteriorates sharply.

Our second finding concerns who within the household bears the cost. In the food arm, women’s dietary diversity declines significantly when rations are reduced, while chil-

dren’s nutritional indicators remain statistically unchanged. This asymmetry is consistent with intra-household allocation norms that prioritise children’s feeding at the expense of women’s dietary quality. This finding implies that household-level indicators alone may understate the welfare costs of transfer reductions, and that complementary measures to protect women’s dietary intake deserve consideration.

Third, cash transfers significantly outperform in-kind assistance. The triple difference-in-differences estimates show that cash yields higher FCS (+0.45 SD), dietary diversity (+0.96 SD), food expenditure (+0.25 SD), and total expenditure (+0.22 SD) relative to in-kind food. The fungibility of cash allows households to allocate resources more efficiently across needs, generating welfare gains that are concentrated in food security and flexible spending. These results reinforce the case for cash-based programming, particularly in the DRC, where the majority of humanitarian assistance remains in-kind.

Fourth, non-uniform in-kind transfers generate community tensions that non-uniform cash transfers do not. In villages where some households received full rations and others received reduced rations, the perception that WFP aid increased conflict rose by 14 percentage points. No equivalent effect is observed in cash villages, and the Social Cohesion Index shows no significant change under non-uniform cash transfers. The visibility of in-kind distributions appears to be the critical mechanism: food is distributed publicly in quantities that neighbours can observe, while cash is received privately. Qualitative evidence reinforces this interpretation—focus group participants described differentiated rations as discriminatory, and resistance was especially acute among women’s groups and in ethnically diverse communities. This finding carries a direct implication: when agencies must introduce non-uniform transfers, delivering them in cash rather than in-kind can substantially mitigate the social costs. Relative to removing beneficiaries entirely—an approach shown to impose significant costs on social cohesion (Olivier et al., 2025)—calibrated reductions delivered in cash may represent a less disruptive path for managing budget constraints.

Fifth, adjusting transfer values for household economies of scale yields no significant welfare losses across any household size category, while generating cost savings of approx-

imately 2.2% of total program expenditure. Even under conservative parameterisations, the savings are positive and could finance the horizontal expansion of coverage to additional vulnerable households. This represents a feasible, low-risk design margin that does not require reducing the quality of support provided to current beneficiaries.

Taken together, these results suggest that when aid budgets contract, the design of the reduction matters as much as its magnitude. Calibrating transfers to estimated need rather than eliminating households preserves coverage, but the welfare costs are substantial and fall disproportionately on women. Three design choices can limit the damage: delivering differentiated transfers in cash rather than in-kind, incorporating equivalence-scale adjustments to free resources for horizontal expansion, and pairing transfer reductions with measures to protect women’s dietary quality.

Several limitations should be noted. The study relies on a six-month follow-up; longer-term effects on human capital, child development, and community dynamics may differ. The cross-modality comparison exploits site-level variation rather than experimental assignment across modalities, and while we assess baseline balance and employ doubly robust methods, unobserved confounders cannot be fully excluded. Finally, the qualitative evidence underscores that the social acceptability of differentiated transfers is highly context-dependent: the strong resistance observed in Kasaï and the concerns raised about conflict-affected eastern provinces suggest that conflict sensitivity assessments should precede any scale-up of this approach.

References

- Aker, J. (2017). Cash or coupons? Testing the impacts of cash versus vouchers in the Democratic Republic of Congo.
- Anderson, M. L. (2008). Multiple inference and gender differences in the effects of early intervention: A reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects. *Journal of the American Statistical Association*, 103(484), 1481–1495.
- Balboni, C., Bandiera, O., Burgess, R., Ghatak, M., & Heil, A. (2022). Why do people stay poor? *Quarterly Journal of Economics*, 137(2), 785–844.
- Banerjee, A., Duflo, E., Goldberg, N., Karlan, D., Osei, R., Parienté, W., Shapiro, J., Thuysbaert, B., & Udry, C. (2019). A multifaceted program causes lasting progress for the very poor: Evidence from six countries. *Science*.
- Bastagli, F., Hagen-Zanker, J., Harman, L., Barca, V., Sturge, G., Schmidt, T., & Pellerano, L. (2016). *Cash transfers: What does the evidence say? A rigorous review of programme impact and of the role of design and implementation features*. Overseas Development Institute.
- Becker, G. S. (1981). *A Treatise on the Family*. Harvard University Press.
- Bonilla, J., Zarzur, R. C., Handa, S., Nowlin, C., Peterman, A., Ring, H., & Seidenfeld, D. (2017). Cash for women’s empowerment? A mixed-methods evaluation of the government of Zambia’s child grant program. *World Development*, 95, 55–72.
- Brune, L., Giné, X., Goldberg, J., & Yang, D. (2016). Facilitating savings for agriculture: Field experimental evidence from Malawi. *Economic Development and Cultural Change*, 64(2), 187–220.
- Chiappori, P.-A. (1992). Collective labor supply and welfare. *Journal of Political Economy*, 100(3), 437–467.

- Crost, B., Felter, J., & Johnston, P. (2014). Aid under fire: Development projects and civil conflict. *American Economic Review*, 104(6), 1833–1856.
- Crosta, A., et al. (2023). Transfer size and welfare outcomes: A review.
- Crosta, A., Gentilini, U., & Valleriani, G. (2024). Transfer size and impacts: An updated review and meta-analysis. World Bank Policy Research Working Paper.
- Deaton, A. & Paxson, C. (1998). Economies of scale, household size, and the demand for food. *Journal of Political Economy*, 106(5), 897–930.
- Della Guardia, A., Gassman, F., Maurer, S., & Pellerano, L. (2022). The effect of cash transfers on community tensions.
- Dustmann, C., Landersø, R., & Andersen, L. H. (2024). Effects of discontinuing unconditional cash transfers.
- Fehr, E. & Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3), 817–868.
- Filmer, D., & Schady, N. (2014). Getting girls into school: Evidence from a scholarship program in Cambodia. *Economic Development and Cultural Change*, 56(3), 581–617.
- Fink, G., Jack, B. K., & Masiye, F. (2020). Seasonal liquidity, rural labor markets, and agricultural production. *American Economic Review*, 110(11), 3351–3392.
- Gentilini, U. (2016). Revisiting the “cash versus food” debate: New evidence for an old puzzle? *World Bank Research Observer*, 31(1), 135–167.
- Gentilini, U., Honorati, M., & Yemtsov, R. (2017). *The state of social safety nets 2017*. World Bank.
- GRFC. (2024). Global Report on Food Crises.
- IPC. (2025). Integrated Food Security Phase Classification: DRC.

- Kondylis, F., & Loeser, J. (2021). Intervention size and persistence. World Bank.
- Lewbel, A. (1997). Consumer demand systems and household equivalence scales. In *Handbook of Applied Econometrics*, Vol. II.
- McIntosh, C., & Zeitlin, A. (2021). Using household grants to benchmark the cost effectiveness of a USAID workforce readiness program. *Journal of Development Economics*, 149, 102582.
- OCHA. (2024). Global Humanitarian Overview.
- Olivier, J., et al. (2025). The social cohesion costs of beneficiary removal in humanitarian programs.
- Paler, L., Prichard, W., Sanchez de la Sierra, R., & Samii, C. (2019). Survey on the effects of aid targeting.
- Premand, P. & Schnitzer, P. (2021). Efficiency, legitimacy, and impacts of targeting methods.
- Quattrochi, J., et al. (2020). Cash transfers in DRC.
- Sant’Anna, P. H. C., & Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1), 101–122.
- Shapiro, J. & Haushofer, J. (2016). The short-term impact of unconditional cash transfers to the poor.
- UNDP. (2024). Human Development Report.
- Verme, P. & Gigliarano, C. (2019). Optimal targeting under budget constraints in a humanitarian context. *World Development*, 119, 224–233.
- WFP. (2023a, 2023b, 2023c). Simulation studies on aid discontinuation.

A Additional Tables and Figures

Table A1: Baseline Balance: In-Kind Food Arm

	Control (uniform ration) 555g/day (1)	Treatment (differentiated) 555g or ~333g/day (2)	(1)–(2)	p-val
Female-headed household	0.39 (0.49)	0.33 (0.47)	-0.06**	0.04
Age of household head	40.79 (14.16)	40.24 (13.40)	-0.55	0.46
Household size	5.72 (2.46)	5.79 (2.31)	0.07	0.66
Children under 5	1.12 (1.14)	1.08 (1.11)	-0.03	0.78
Elderly members	0.18 (0.46)	0.17 (0.44)	-0.01	0.59
Pregnant or lactating women	0.43 (0.50)	0.45 (0.50)	0.02	0.60
Head has no education	0.38 (0.49)	0.44 (0.50)	0.06	0.50
Resident household	0.72 (0.45)	0.75 (0.43)	0.03	0.74
Food Consumption Score	27.69 (9.45)	28.10 (10.75)	0.41	0.69
Reduced Coping Strategy Index	12.98 (10.03)	13.65 (10.43)	0.66	0.47
Household Dietary Diversity	3.66 (1.17)	3.53 (1.23)	-0.13	0.34
Food expenditure (ln)	11.40 (0.70)	11.34 (0.66)	-0.07	0.26
Non-food expenditure (ln)	10.26 (1.06)	10.26 (1.10)	-0.00	0.99
Total expenditure (ln)	11.75 (0.69)	11.70 (0.67)	-0.05	0.47
Food expenditure share	72.65 (15.34)	71.30 (15.53)	-1.35	0.30
Food insecure (binary)	0.89 (0.31)	0.87 (0.33)	-0.02	0.52
Livelihood coping (crisis/emerg.)	0.52 (0.50)	0.58 (0.49)	0.05*	0.10
Social cohesion index	-0.08 (1.98)	0.06 (2.07)	0.14	0.46
Conflict in past 30 days	0.18 (0.38)	0.24 (0.43)	0.06	0.37
Observations	1391	1361		

Notes: Full ration corresponds to the standard food basket (555g/person/day). Calibrated ration is approximately 40% lower, determined by a needs-based assessment of household food expenditure capacity. Difference column reports OLS coefficient with standard errors clustered at the village level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Baseline Balance: Cash Arm

	USD 14 (1)	USD 18 (2)	EE (3)	F-test p-val
Female-headed household	0.28 (0.45)	0.25 (0.44)	0.28 (0.45)	0.69
Age of household head	44.28 (14.52)	46.33 (14.25)	44.54 (13.96)	0.07*
Household size	5.47 (2.56)	5.52 (2.41)	5.97 (2.53)	0.12
Children under 5	1.08 (1.06)	1.03 (1.02)	1.07 (1.00)	0.80
Elderly members	0.20 (0.47)	0.26 (0.54)	0.19 (0.45)	0.13
Pregnant or lactating women	0.46 (0.50)	0.45 (0.50)	0.49 (0.50)	0.54
Head has no education	0.30 (0.46)	0.26 (0.44)	0.20 (0.40)	0.00***
Resident household	0.97 (0.17)	0.99 (0.10)	0.98 (0.14)	0.09*
Food Consumption Score	27.36 (10.78)	28.20 (12.34)	30.09 (12.47)	0.27
Reduced Coping Strategy Index	18.11 (12.18)	18.16 (12.47)	16.87 (12.50)	0.72
Household Dietary Diversity	4.01 (1.08)	4.03 (1.08)	4.20 (1.13)	0.22
Food expenditure (ln)	11.22 (0.66)	11.34 (0.67)	11.33 (0.64)	0.28
Non-food expenditure (ln)	10.18 (1.03)	10.25 (1.14)	10.43 (0.90)	0.06*
Total expenditure (ln)	11.58 (0.68)	11.70 (0.70)	11.72 (0.64)	0.23
Food expenditure share	71.29 (14.33)	71.55 (15.36)	68.96 (13.68)	0.09*
Food insecure (binary)	0.89 (0.31)	0.87 (0.34)	0.83 (0.38)	0.34
Livelihood coping (crisis/emerg.)	0.68 (0.47)	0.73 (0.45)	0.69 (0.46)	0.41
Social cohesion index	-0.12 (1.82)	0.01 (1.93)	0.11 (1.71)	0.41
Conflict in past 30 days	0.02 (0.14)	0.01 (0.09)	0.02 (0.15)	0.10
Observations	800	940	793	

Notes: USD 14 covers food needs only. USD 18 adds a non-food component (USD 4/person). EE adjusts the transfer value for economies of scale by household size. *F*-test reports the *p*-value from a joint test of equality across all three groups, with standard errors clustered at the village level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Baseline Balance: In-Kind (Full Ration) vs. Cash (USD 14)

	In-Kind (Full ration) (1)	Cash (USD 14) (2)	(1)–(2)	p-val
Female-headed household	0.39 (0.49)	0.28 (0.45)	-0.11***	0.00
Age of household head	40.79 (14.16)	44.28 (14.52)	3.49***	0.00
Household size	5.72 (2.46)	5.47 (2.56)	-0.24**	0.03
Children under 5	1.12 (1.14)	1.08 (1.06)	-0.04	0.45
Elderly members	0.18 (0.46)	0.20 (0.47)	0.02	0.28
Pregnant or lactating women	0.43 (0.50)	0.46 (0.50)	0.03	0.18
Head has no education	0.38 (0.49)	0.30 (0.46)	-0.08***	0.00
Resident household	0.72 (0.45)	0.97 (0.17)	0.26***	0.00
Food Consumption Score	27.69 (9.45)	27.36 (10.78)	-0.34	0.46
Reduced Coping Strategy Index	12.98 (10.03)	18.11 (12.18)	5.12***	0.00
Household Dietary Diversity	3.66 (1.17)	4.01 (1.08)	0.34***	0.00
Food expenditure (ln)	11.40 (0.70)	11.22 (0.66)	-0.18***	0.00
Non-food expenditure (ln)	10.26 (1.06)	10.18 (1.03)	-0.08*	0.09
Total expenditure (ln)	11.75 (0.69)	11.58 (0.68)	-0.17***	0.00
Food expenditure share	72.65 (15.34)	71.29 (14.33)	-1.36**	0.04
Food insecure (binary)	0.89 (0.31)	0.89 (0.31)	-0.00	0.93
Livelihood coping (crisis/emerg.)	0.52 (0.50)	0.68 (0.47)	0.15***	0.00
Social cohesion index	-0.08 (1.98)	-0.12 (1.82)	-0.04	0.62
Conflict in past 30 days	0.18 (0.38)	0.02 (0.14)	-0.15***	0.00
Observations	1391	800		

Notes: Compares households receiving the full in-kind ration with those receiving the USD 14 cash transfer (food component only). Difference column reports OLS coefficient with robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Expenditure Decomposition: Food Groups and Non-Food Items (Food Arm)

	(1) OLS DiD	(2) Ctrl Mean (BL)	(3) % Change
<i>Panel A: Aggregate expenditure</i>			
Food Expenditure - ln	-0.37*** (0.04)	11.23	-30.9%
Non Food Expenditure - ln	-0.44*** (0.07)	10.05	-35.6%
Total Expenditure - ln	-0.39*** (0.04)	11.57	-32.3%
<i>Panel B: Food group expenditure (ln)</i>			
Expenditure on cereals - ln	-1.71*** (0.46)	7.13	-81.9%
Expenditure on tubers - ln	0.15 (0.13)	9.85	16.2%
Expenditure on pulses - ln	-2.64*** (0.50)	3.77	-92.9%
Expenditure on vegetables - ln	-0.01 (0.11)	9.41	-1.0%
Expenditure on fruits - ln	-2.77*** (0.35)	2.46	-93.7%
Expenditure on meat/fish - ln	-4.53*** (0.30)	3.54	-98.9%
Expenditure on oils/fats - ln	-3.15*** (0.36)	5.87	-95.7%
Expenditure on dairy - ln	-0.69*** (0.14)	0.11	-49.8%
<i>Panel C: Non-food expenditure (ln)</i>			
Expenditure on health - ln	-0.26 (0.23)	8.67	-22.9%
Expenditure on education - ln	-1.12*** (0.29)	3.94	-67.4%
Expenditure on debt repayment - ln	-0.74* (0.43)	3.68	-52.3%
Savings set aside - ln	-0.76*** (0.19)	0.54	-53.2%
Controls	Yes		

Notes: OLS DiD estimates for the food arm. All expenditure variables are in log form ($\ln(\text{expenditure} + 1)$). The % Change column reports the exact percentage change: $(e^{\hat{\beta}} - 1) \times 100$. Panel A shows aggregate categories; Panel B decomposes food expenditure by food group; Panel C shows non-food expenditure categories. Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Food Consumption Score Groups: Poor, Borderline, and Acceptable (Food Arm)

	(1) OLS	(2) DR-DiD	(3) IPW
FCS: Poor (<28)	1.09*** (0.03)	1.10*** (0.03)	1.13 (0.00)
FCS: Borderline (28–42)	-0.80*** (0.03)	-0.81*** (0.03)	-0.82 (0.00)
FCS: Acceptable (>42)	-0.29*** (0.02)	-0.29*** (0.02)	-0.30 (0.00)
Controls	Yes	Yes	No
Weights	None	DR	IPW
Observations	5524	5524	5524

Notes: Each row reports the DiD coefficient for a dummy indicating FCS category membership: poor (< 28), borderline (28–42), or acceptable (> 42). A positive coefficient on “poor” indicates an increase in the share of households falling into that category. Same specifications as Table 3. Standard errors clustered at the village level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Heterogeneous Effects of Needs-Based Transfer Calibration (Food Arm)

	By Baseline Vulnerability			By HH Size			By Head Gender	
	Most vuln. (1)	Medium (2)	Least vuln. (3)	Small (4)	Medium (5)	Large (6)	Female (7)	Male (8)
Food Consumption Score	0.00 (0.00)	-15.37*** (1.01)	-17.25*** (0.95)	-17.23*** (1.03)	-18.54*** (0.91)	-17.93*** (0.82)	-17.79*** (0.83)	-18.25*** (0.88)
Household Dietary Diversity	0.00 (0.00)	-1.86*** (0.14)	-1.83*** (0.13)	-1.76*** (0.12)	-2.07*** (0.12)	-2.02*** (0.12)	-1.97*** (0.12)	-1.99*** (0.11)
Reduced Coping Strategy	0.00 (0.00)	-1.24 (1.51)	1.34 (1.11)	-0.18 (1.08)	1.10 (1.08)	0.88 (1.26)	-0.54 (0.94)	1.55* (0.90)
Food Expenditure - ln	0.00 (0.00)	-0.46*** (0.08)	-0.35*** (0.04)	-0.32*** (0.06)	-0.39*** (0.05)	-0.41*** (0.06)	-0.32*** (0.06)	-0.41*** (0.04)
Total Expenditure - ln	0.00 (0.00)	-0.36*** (0.09)	-0.41*** (0.04)	-0.29*** (0.06)	-0.41*** (0.05)	-0.43*** (0.06)	-0.34*** (0.05)	-0.42*** (0.04)
Share of Food Insecure HHs	0.00 (0.00)	0.24*** (0.06)	0.31*** (0.04)	0.27*** (0.04)	0.37*** (0.04)	0.37*** (0.05)	0.32*** (0.03)	0.36*** (0.04)
Minimum Acceptable Diet (Children)	0.00 (0.00)	-0.02 (0.02)	-0.01 (0.01)	0.00 (0.02)	-0.02 (0.02)	-0.03* (0.02)	-0.03 (0.02)	-0.01 (0.01)
Minimum Dietary Diversity (Women)	0.00 (0.00)	-0.63*** (0.20)	-0.39** (0.15)	-0.42* (0.22)	-0.64*** (0.14)	-0.61*** (0.19)	-0.58*** (0.17)	-0.59*** (0.14)
Controls	Yes							

Notes: Columns (1)–(3) interact the DiD treatment effect with baseline FCS terciles (most vulnerable is the reference group, which received the full ration). Columns (4)–(6) interact with household size categories. Columns (7)–(8) interact with household head gender. All specifications include baseline controls and cluster standard errors at the village level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Social Cohesion: Anderson (2008) Index and Components — Cash Arm

	(1) Coeff.	(2) Ctrl Mean	(3) Ctrl SD
<i>Summary index</i>			
Social Cohesion Index (Anderson 2008)	0.06 (0.12)	-0.04	0.90
<i>Individual components</i>			
Relations between groups	-0.13 (0.10)	1.92	0.74
Active relations between groups	-0.06 (0.05)	1.80	0.54
Frequency of contact	0.10 (0.08)	2.25	0.84
Quality of interactions	0.06 (0.08)	2.36	0.81
Frequency of interactions	0.08 (0.10)	2.51	0.87
Trust	0.04 (0.07)	2.12	0.68
Sense of belonging	0.04 (0.08)	1.92	0.74
Voluntary participation	0.10 (0.11)	2.84	1.06
Information sharing	0.31** (0.14)	3.14	1.14
Controls	Yes		

Notes: ITT estimates for the cash arm (endline only), exploiting FDP-level randomization. The summary index is constructed using the Anderson (2008) inverse-covariance-weighted approach, standardized to mean 0 and SD 1 using the control group. Individual components are the 9 social cohesion items. Treatment is assignment to non-uniform transfer values. Standard errors clustered at the village level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.